

Master Chord Engine v1.2.0: Operations Manual

This Logic Pro Scripter layout transforms any compact MIDI keyboard into an interactive chord design engine. Your left hand commands chord geometry, inversion voicing, and layout zones via a customizable **Control Zone**, while your right hand triggers harmonized chord structures across the **Performance Zone** using mathematically mirrored release timing to eliminate stuck notes.

1. The Interactive Dashboard UI

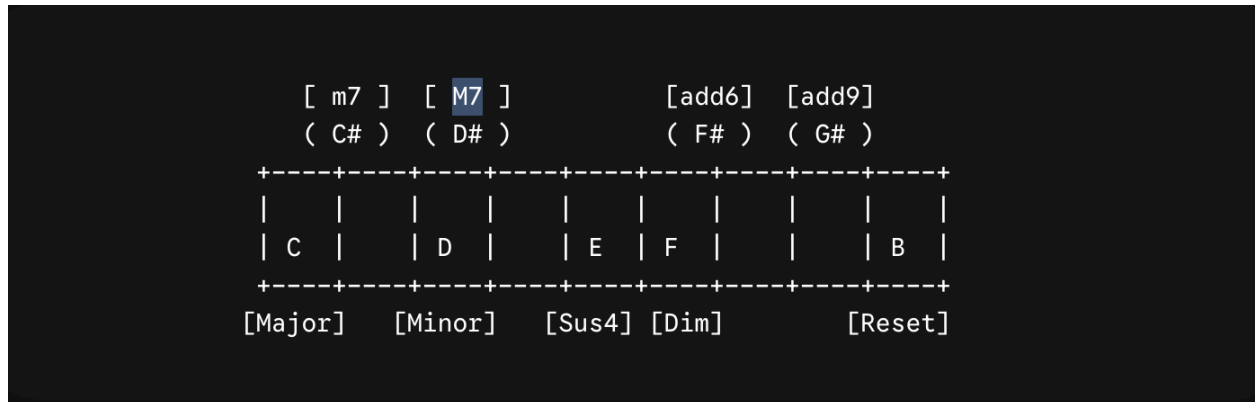
When the script is running inside Logic Pro's Scripter plugin, the parameter dashboard displays six real-time configuration controls:

- **Control Octave (Dropdown: C0 / C1 / C2 / C3):** Shifts the entire 12-key left-hand modifier map to the octave that best matches your physical setup. *Settings Tip:* Default is **C2**. Lower it to **C0** or **C1** on a compact 37-key keyboard to keep the modifier console tucked completely out of the way of your right-hand performance keys.
- **Triad Voicing (Dropdown: Root Position / 1st Inversion / 2nd Inversion):** * *Root Position:* Spreads your chords with the fundamental root note sitting flat at the bottom (1 - 3 - 5).
 - *1st Inversion:* Inverts the fundamental root note up an octave, leaving the structural 3rd voice sitting at the base of the chord layout (3 - 5 - 1). Great for smoothing out step-wise vocal transitions.
 - *2nd Inversion:* Shifts both your fundamental root note and your 3rd up an octave scale, pinning the solid 5th interval directly to the floor of your chord stack (5 - 1 - 3). Excellent for thick, swelling keyboard pad tracking.
- **Auto-Bass Register (Dropdown: OFF / C0 Range / C1 Range / C2 Range):** * *OFF:* Disables the bass layer entirely.
 - *C0 Range (Sub):* Generates a parallel sub-bass note locked below MIDI 24 (C1 and lower) for deep, sub-woofer heavy weight.
 - *C1 Range (Low - Default):* Locks the bass note into the traditional 24-35 register for clean, standard synth bass tracking.
 - *C2 Range (Mid):* Elevates the bass line into the 36-47 middle pocket—ideal for creating tight walking-bass counter melodies.
- **Strum Engine (Dropdown: OFF / ON):**
 - *ON:* Staggers the note delivery departures sequentially to mimic an organic string roll or harp sweep.
 - *OFF:* Fires all chord voices simultaneously for sharp, immediate rhythmic blocks.
- **Strum Direction (Dropdown: Low to High / High to Low):**
 - *Low to High (Up):* Arpeggiation rolls upwards from the base octave to the highest extension key, perfectly emulating a standard guitar upward stroke.

- *High to Low (Down)*: Flips the sorting matrix completely. Notes fire from the absolute ceiling of your chord cluster down to the fundamental performance root note, mimicking a cascading harp sweep or an intentional pianistic down-stroke.
- **Strum Speed (ms) (Linear Slider: 0 to 360 ms)**: Adjusts the delay time between each notes' arrival when the Strum Engine is active. Keep it at **15–30ms** for crisp guitar-style sweeps, or lift it to **150–360ms** for cascading ambient arpeggios.
- **Strum Size (Dropdown: 1 Octave / 2 Octaves / 3 Octaves)**: Scales your active chord configuration vertically up the keyboard ladder. At 3 octaves with active extensions, a single right-hand key down event generates up to 15 layered note streams.

2. Left-Hand Controller Blueprint

Set your **Control Octave** parameter to match your desired location. The modifier console maps relative white-key and black-key triggers across a clean 12-note layout starting from your chosen root key (C):



Triad Foundations (White Keys)

Tap once to latch a foundation. Switching to any new white key automatically resets all black-key extensions to clean your sonic slate.

- **C (Offset 0)**: Major Triad (1 - 3 - 5)
- **D (Offset 2)**: Minor Triad (1 - \flat3 - 5)
- **E (Offset 4)**: Suspended Triad (1 - 4 - 5)
- **F (Offset 5)**: Diminished Triad (1 - \flat3 - \flat5)

Extension Toggles (Black Keys)

Tap once to engage an extension; tap a second time to turn it off.

- **C# (Offset 1)**: Minor 7th (m7) — *Disables M7 automatically if active.*
- **D# (Offset 3)**: Major 7th (M7) — *Disables m7 automatically if active.*
- **F# (Offset 6)**: add6 Extension
- **G# (Offset 8)**: add9 Extension (*Voiced one octave up for maximum clarity.*)

Quick Reset Switch

- **B (Offset 11):** Clears all active extensions back to a completely dry triad layout without altering your current foundation type.

3. Symmetrical Chronological Tracking

The core engine features a robust, tracking-independent memory pipeline (**activePitchesMemory**). Every generated pitch is logged directly alongside its absolute index location (**delayIndex**) in the strum queue the exact microsecond a key is pressed.

When your hand releases a performance note, the script ignores any adjustments made to your parameter sliders, inversions, or left-hand modifiers mid-phrase. Instead, it reads the memory log directly to dispatch mathematically mirrored, time-delayed **NoteOff** events. This fully matches your strum timeline, ensuring that every note-on event receives a perfect shutoff command regardless of performance velocity.

💡 **Voice Order Sorting:** Notice that when you engage an inversion or switch directions with the **Strum Engine** active, the script automatically recalculates the ascending or descending pitch sequence. Your strum arpeggio will organically roll using the *new* layout order, keeping your performance voice transitions incredibly natural.

4. First-Time Setup & Preset Saving

Step 1: Insert the Scriptor Plugin

1. Create or select a **Software Instrument** track inside Logic Pro.
2. In the channel strip, locate the **MIDI FX** slot directly above your instrument plugin slot (e.g., Alchemy, Retro Synth, or ES2).
3. Click the right edge of the MIDI FX slot and select **Scripter** from the drop-down menu.

Step 2: Inject the Production Code

1. Once the Scriptor interface displays, click the **Editor** button in the top-left corner to reveal the JavaScript text window.
2. Select all placeholder text inside the panel (**Cmd + A**) and hit **Delete**.
3. Copy the entire **v1.2.0 Master Edition** script block and paste it directly into the empty editor (**Cmd + V**).
4. Click the **Run Script** button in the top-right corner of the window. The code viewer will close, and your interactive parameter dashboard layout will deploy.

Step 3: Save Your Custom User Preset

1. Click the **Preset drop-down menu** at the very top center of the Scriptor plugin wrapper interface (defaults to "Factory Default").
2. Select **Save As...** from the bottom of the window menu options.
3. Name the patch file **Chord Engine v1.2.0** and click **Save**.

The blueprint is now permanently stored in your local Logic preferences. For all future projects, simply load a blank Scriptor plugin and select your preset from the drop-down list to recall your entire performance routing matrix instantly.

5. Capturing Chords via "Record MIDI Here"

By default, when you press Record in Logic Pro, the DAW captures the raw, literal notes coming off your keyboard hardware *before* they enter the MIDI FX slot. This means your recorded MIDI region will only contain single right-hand root notes and left-hand modifier command keys.

To bypass this and write the fully generated chords directly into your piano roll in real-time, you use Logic's native **Record MIDI Here** processing path.

Why It Is Needed

- **Instant Editing:** It automatically converts the script's real-time math into multi-note chord blocks, allowing you to instantly quantize, humanize velocity, or adjust individual chord extensions in the piano roll the second you hit stop.
- **Arrangement Freedom:** Once the chords are printed as permanent MIDI notes, you can safely copy, loop, or drag that MIDI region to *any other instrument track* in your project—even tracks without the Scriptor plugin loaded.

How to Enable It

1. Open the channel strip for your active Software Instrument track.
2. Locate the **MIDI FX** slot where your script is currently running.
3. Right-click (**Ctrl + click**) or click the drop-down chevron on the Scriptor slot to open its contextual menu.
4. Select **Record MIDI Here** from the list. The slot will update with a small target marker icon, signaling that the recorder playhead has been moved to the output stage of the plugin.

What the Output Will Look Like

When you hit record and play a phrase, you will visually see standard, single-note regions being drawn on the timeline. However, the moment you press the spacebar to stop recording, Logic flushes the script's calculations into the region.

Opening the **Piano Roll (P)** will reveal the transformed output:

- **The Command Notes Are Gone:** The left-hand foundation switch keys (like your low command notes) are automatically filtered out and erased so they don't trigger unwanted low-end synth frequencies.
- **Full Chord Blocks:** Your single right-hand root notes will be replaced by dense vertical stacks of notes representing your major, minor, or extended chords.
- **Staggered Strum Visuals:** If the **Strum Engine** was active during the take, you will see the notes inside the piano roll stepping upward or downward like a microscopic

staircase, capturing your exact millisecond strum speed settings with perfect timeline precision.